

SENTINEL AI SURVEILLANCE PLATFORM

Unified CCTV Ingestion, AI ANPR & GIS Vehicle Tracking System

An enterprise-grade, 100% self-hosted computer vision intelligence system. Integrates multi-vendor cameras, executes sub-second hotlist alerts, and reconstructs vehicle journeys on interactive GIS maps with **zero cloud API costs**.

Multi-Vendor RTSP/ONVIF

YOLOv8 + PP-OCRv4

<400ms Hotlist Siren

PostGIS Breadcrumbs

The Surveillance Challenge Faced by Police Forces

Current city surveillance systems suffer from fragmented hardware, slow alerting, and high cloud costs.



Multi-Vendor Silos

Cities have thousands of cameras from different brands (Hikvision, Dahua, CP Plus, Axis) running incompatible proprietary VMS software.

✗ No unified live video matrix across zones

✗ Difficult to add or swap camera brands



Complex Indian Plates

Standard foreign OCR systems fail on Indian roads due to high font variations, 2-line plates, commercial yellow plates, and dust/glare.

✗ 40%+ OCR errors on Indian standard fonts

✗ Night-vision headlight glare failures



Delayed Suspect Intercept

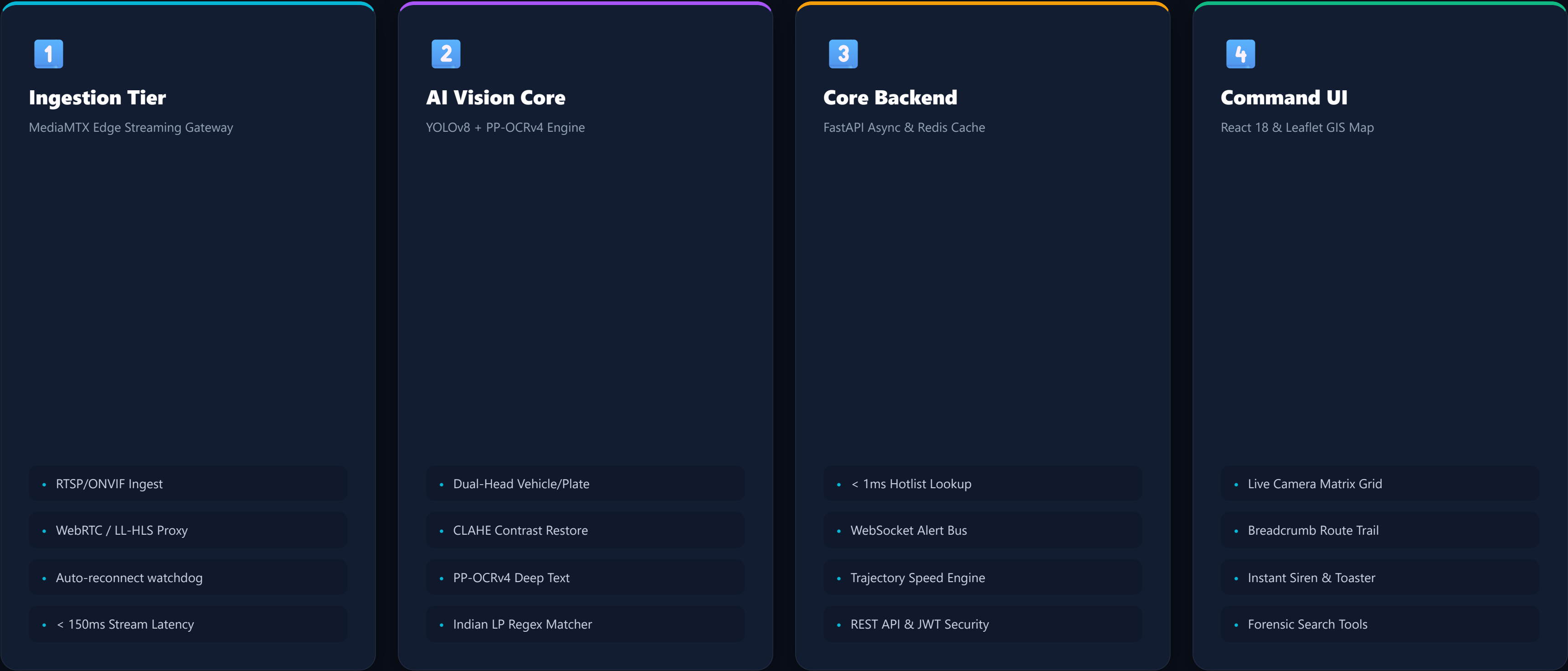
Stolen vehicles and wanted criminals cross checkpoints unnoticed because manual monitoring cannot check hundreds of streams in real time.

✗ Hours lost in manual DVR playback

✗ Exorbitant per-camera cloud license fees

High-Level Architecture & Component Stack

Modular 5-tier architecture built for non-blocking asynchronous processing and extreme scalability.



Specialized AI ANPR Pipeline for Indian Conditions

Multi-stage pipeline engineered specifically for high accuracy under difficult lighting, angles, and plate formats.

⚡ 5-Step Deep Learning Pipeline

Executes in under 100ms per frame entirely on local hardware

- 1. Adaptive Sampler:** Motion-triggered frame grabbing throttles to 10 FPS to save compute.
- 2. Dual-Head YOLOv8:** Isolates vehicle class (Car, Bike, Truck, Auto) and crops the license plate rectangle.
- 3. CLAHE Image Enhancement:** Normalizes night headlight glare, shadow contrast, and perspective deskew.
- 4. PaddleOCR (PP-OCRv4):** Character recognition trained on Indian multi-font, HSRP, and 2-line plates.
- 5. Regex Validation:** Enforces state/RTO syntax & corrects optical ambiguities (0 vs O, 8 vs B).

📺 Live Detection Simulation & Benchmarks

Validated on real-world Indian highway feeds

CAMERA: SG Highway (CAM-04)

• INFERENCE: 42ms (GPU)

GJ 01 AB 1234

VEHICLE: WHITE FORTUNER (SUV) • CONFIDENCE: 96.4%

97.4%

Detection Accuracy

< 45ms

GPU Frame Latency

Sub-Second Watchlist Matching & Operator Siren

Instant broadcast from camera sensor to operator workstation in under 400 milliseconds.

**Real-Time Hotlist Match Engine**

Integrates with CCTNS & police stolen vehicle databases

Redis In-Memory Key-Value: Performs hash lookups in <1ms per recognized plate.

Fuzzy Levenshtein Matcher: Tolerates single-character OCR scratches to catch disguised plates.

Severity Categorization: Distinguishes CRITICAL (Stolen/Armed), HIGH (Wanted), and MEDIUM alerts.

Instant Audio Siren: Loud sound chime and red flashing toaster alerts the operator immediately.

**Command Center Alert Card Example**

What the police operator sees in real time:

**CRITICAL HOTLIST MATCH**

FIR-2026-9081 • STOLEN SUV REPORTED

< 380ms

**Location:** SG Highway - Pakwan Junction (CAM-04)

**Timestamp:** Today at 10:45:22 AM

**Vehicle:** GJ01AB1234 (White Toyota Fortuner)

**Police Station:** Vastrapur PS, Ahmedabad

1-Click Live Feed

Dispatch PCR Van

Interactive GIS Map & Breadcrumb Trajectory Replay

Reconstruct suspect escape routes with spatial speed estimation and directional polylines.



Live GIS Overview

Interactive Leaflet & OpenStreetMap map displaying all registered city CCTV nodes with live status pins.

- Green/Red camera health pins
- Flashing radar alert marker
- Zero Google Maps licensing fees



Numbered Breadcrumbs

Enter any license plate number to trace the chronological path taken across camera junctions (1 → 2 → 3 → 4).

- Directional route polylines
- Speed calculation (km/h)
- Delta-T transit duration



Forensic Dossier

Export court-admissible PDF investigation dossiers containing full trajectory history and high-res vehicle crops.

- 1-Click PDF evidence export
- Full-frame snapshots + plate crops
- Wildcard search ('GJ01??1234')

Unified Web Command Center Capabilities

Modern React 18 interface designed specifically for fast-paced police control room operations.

 **Live Multi-Cam Grid**

1, 4, 9, or 16-channel low-latency WebRTC streams with AI bounding box overlays.

 **Alert History & Logs**

Real-time alert feed with 1-click acknowledgement and operator audit logs.

 **Vehicle Search**

Search by license plate, camera location, vehicle class, and custom date/time windows.

 **Camera & Hotlist Hub**

Bulk CSV import for CCTNS FIR lists and RTSP camera registration management.

< 150ms

WebRTC Stream Latency

100% Async

Non-Blocking ASGI

Air-Gapped

Zero External APIs

CI/CD Ready

GitHub Actions Automated

Summary, Operational ROI & Next Steps

High operational impact with massive recurring cost savings for government infrastructure.

💰 Massive Cost & Infrastructure ROI

Zero Recurring SaaS Fees: Eliminates lakhs to crores in annual per-camera licensing fees.

Hardware Reusability: Unifies existing legacy Hikvision/Dahua/CP Plus cameras without costly replacements.

100% Data Sovereignty: Video footage and criminal records remain strictly inside the State Data Centre.

Rapid Resolution: Cuts forensic tracking time from several hours down to a few seconds.

🚀 Future Roadmap Enhancements

Section-Control Speed Radar: Automated speed violation detection across highway corridors.

Traffic Violation E-Challan: Helmet, tripling, and red-light jumping detection modules.

Edge AI Checkpoint Boxes: NVIDIA Jetson deployment for remote highway toll plazas.

Inter-District Police Sync: Real-time hotlist sync across Ahmedabad, Surat, and Rajkot zones.